

LAB RESOURCE UTILIZATION PLATFORM

Document 7

Database Schema

Infosys Springboard Internship

1. Introduction

The database schema defines the structure of the Lab Resource Utilization Platform. It stores information related to institutions, departments, users, equipment, bookings, maintenance, utilization, notifications, and resource sharing while maintaining data integrity through primary and foreign keys.

2. Entity Relationship Diagram

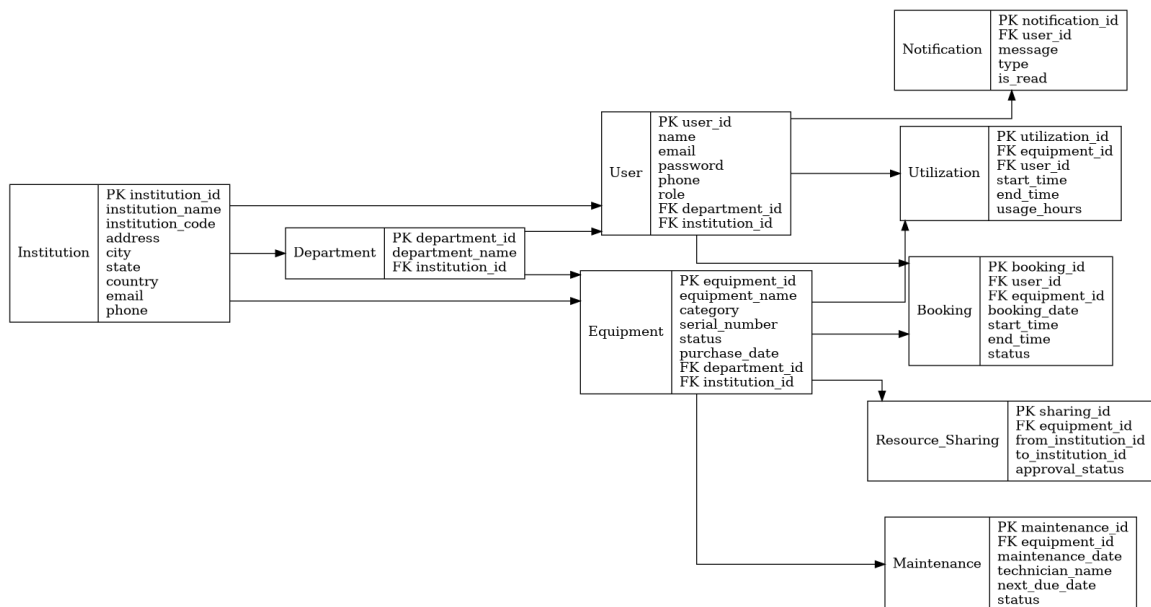


Figure 1: Entity Relationship Diagram of the Lab Resource Utilization Platform

3. Entity Description

Entity	Purpose
Institution	Stores institution details.
Department	Stores department information.
User	Stores user credentials and roles.

Equipment	Stores laboratory equipment details.
Booking	Stores equipment booking records.
Utilization	Tracks equipment usage.
Maintenance	Maintains service history.
Notification	Stores user notifications.
Resource Sharing	Manages equipment sharing between institutions.

4. Relationship Summary

- One Institution can have multiple Departments, Users, and Equipment.
- Each Department contains multiple Users and Equipment.
- Users can create multiple Bookings and Utilization records.
- Equipment can have multiple Bookings, Maintenance records, and Utilization records.
- Notifications are linked to individual Users.
- Resource Sharing connects Equipment between Institutions.

5. Advantages

- Reduces data redundancy.
- Maintains referential integrity.
- Supports efficient queries.
- Improves scalability.
- Simplifies maintenance.

6. Conclusion

The proposed relational database schema provides a secure and efficient foundation for managing laboratory resources and supports future expansion of the platform.